

Semester	Course Code	Name of the course	L	T	P	Credits
V	CS5M004	Ethics, Bias and Safety in LLM	2	0	0	2

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Students will be able to Analyze ethical challenges in LLM development and deployment.
2	CO2	Students will be able to Detect and mitigate bias in training datasets and model outputs.
3	CO3	Students will be able to Assess and improve the safety of LLM-based applications.
4	CO4	Students will be able to Understand legal and regulatory frameworks surrounding AI usage.
5	CO5	Students will be able to Evaluate and propose solutions to ethical controversies in LLMs using real-world case studies.

Unit III	Bias in Training Data and Outputs: Sources of bias: gender, race, culture, Dataset auditing and de-biasing techniques, Evaluation metrics for bias and fairness, Societal impacts of biased outputs.	04
----------	---	----

1. Gender Bias

Gender bias in LLMs arises when models replicate and reinforce societal stereotypes about gender roles or identities. This bias may be due to skewed representation in training data or historical gender norms.

Example:

- The model may associate the word 'nurse' more often with women and 'engineer' with men.
- When generating job recommendations or descriptions, it might suggest technical roles for males and caregiving roles for females.

Racial Bias

Racial bias occurs when LLMs reflect or amplify stereotypes, prejudices, or underrepresentation of certain racial or ethnic groups. This can result in unequal treatment or mischaracterization in generated content.

Example:

- Associating names commonly linked to certain racial groups with criminality or lower socioeconomic status.
- Describing people of color using more negative or stereotyped language compared to white individuals.

Cultural Bias

Cultural bias refers to the dominance of one cultural worldview—often Western—over others in the training data. This can result in the marginalization or misrepresentation of non-Western values, languages, or traditions.

Example:

- The model may lack knowledge or provide inaccurate responses about festivals, practices, or beliefs outside of the Western world.
- Non-English idioms or sayings may be translated poorly or misunderstood due to low-resource language data.

Racism

Racism in AI happens when algorithms show unfair bias against certain racial or ethnic groups. This can lead to harms like wrongful arrests from facial recognition misidentifications or biased hiring algorithms limiting job opportunities. AI often replicates biases in its training data, reinforcing systemic racism and deepening racial inequalities in society.

Examples

Facial recognition software misidentifies certain races, leading to false arrests.

Job recommendation algorithms favor one racial group over another.

AI-driven diagnostic tools for skin cancer are less accurate for individuals with dark skin due to non-diverse training datasets.

White saviour stereotype



Data Auditing

Data auditing is the process of conducting a thorough review and analysis of a company's data to ensure its accuracy, consistency, and security.

This process typically involves checking if the data adheres to internal guidelines and regulations, whether it's complete and consistent, and if proper safeguards are in place.

Data auditing also evaluates data management practices to ensure organizations gather, store, and use data appropriately and effectively. It helps identify any issues or errors in data, which in turn aids in making informed decisions and setting effective business strategies.

Data auditing is needed for several reasons:

Improve Data Quality: Data audits can identify errors, inconsistencies, and redundancies in data sets, enabling organizations to improve the accuracy and quality of their data.

Ensure Compliance: With increasing regulations around data protection and privacy, data audits help ensure compliance by identifying any potential areas of non-compliance and allowing businesses to address them proactively.

Enhance Security: By examining data storage, access, and handling processes, data audits can identify security vulnerabilities and help organizations enhance data security.

Inform Decision-Making: High-quality, reliable data is critical for decision-making. Data audits can ensure that the data businesses are using to make decisions is accurate and valid.

Data auditing is needed for several reasons:

Maximize Resource Use: Daunting can uncover data issues that may be wasting resources. Once identified, these can be addressed to maximize efficiency and effectiveness.

Maintain Customer Trust: Data breaches or incorrect data usage can damage a business's reputation. Regular data audits can mitigate this risk and maintain customer trust.

How to Conduct a Data Audit?

Every organization is unique, so its data audit might look different depending on its particular needs and objectives. Conducting a data audit involves several key steps:

Defining the Objectives: Clearly outline what you hope to achieve from the data audit. This could be anything from checking for compliance to better understanding how your organization uses data.

Identifying Data Sources: Understand where your organization's data comes from. This includes internal sources like databases or CRM systems and external sources like social media or third-party apps.

Mapping the Data Flow: Document how data moves around your organization. This process, often known as data mapping, will help you understand who accesses the data, how it is shared, and how long it's kept.

Evaluating Data Quality: This involves profiling to understand the underlying quality issues afflicting the data. In this process, check for inconsistencies in your data sets, such as errors or duplicates. Also, assess if the data is timely, accurate, relevant, and complete.

Reviewing Security Measures: Ensure your organization has robust security measures to prevent breaches. This also involves assessing the potential impact of poor data quality on security preparedness and performance.

Checking for Compliance: Depending on your industry and location, your organization might need to comply with certain data regulations. Ensure your data management practices align with these laws.

Creating an Audit Report: Compile your findings into a comprehensive report once the data audit is complete. This document should outline your data sources, data quality, security measures, and compliance status. It should also contain recommendations for improvement.

Implementing Changes: Use the recommendations from your audit report to improve your organization's data management practices. This could involve cleaning up your data sets or updating your security systems.

Repeat: Data is never static, so regular audits are needed to ensure it remains high quality, secure, and compliant.

Risks of Not Performing a Data Audit

Not performing a data audit can pose numerous risks, including:

Poor Quality Data: Without regular audits, errors, inaccuracies, and outdated information can exist in the data, leading to faulty insights and unwise business decisions.

Non-Compliance: Data audits help ensure compliance with data protection laws and industry regulations. A lack of auditing could result in violation fines and legal problems.

Security Threats: Data audits often involve assessing file protection methods. Without them, data vulnerabilities may go unnoticed, increasing the risk of data breaches and theft.

Inefficient Operations: Audits can reveal areas where data processes might be redundant or ineffective. Ignoring audits may lead to continued inefficiencies, wasted resources, and missed opportunities.

Loss of Customer Trust: If customer data isn't audited and maintained properly, it can lead to issues that harm the company's reputation, customer trust, and, ultimately, customer relationships.

Poor Decision Making: A data audit assures the data's quality, consistency, and validity on which the strategic decisions are based. The absence of an audit may lead to poor decision-making.

Inadequate Data Governance: Without data audits, it can be difficult to implement effective data governance strategies, as companies may lack a complete and accurate understanding of their data landscape.

Sources of bias: gender, race, culture



J D College of Engineering
& Management, Nagpur
An Autonomous Institute

Sources of bias: gender, race, culture



J D College of Engineering
& Management, Nagpur
An Autonomous Institute

Sources of bias: gender, race, culture



J D College of Engineering
& Management, Nagpur
An Autonomous Institute

Sources of bias: gender, race, culture



J D College of Engineering
& Management, Nagpur
An Autonomous Institute

Sources of bias: gender, race, culture



J D College of Engineering
& Management, Nagpur
An Autonomous Institute

Sources of bias: gender, race, culture



J D College of Engineering
& Management, Nagpur
An Autonomous Institute

Sources of bias: gender, race, culture



J D College of Engineering
& Management, Nagpur
An Autonomous Institute

Sources of bias: gender, race, culture



J D College of Engineering
& Management, Nagpur
An Autonomous Institute

Sources of bias: gender, race, culture



J D College of Engineering
& Management, Nagpur
An Autonomous Institute

Sources of bias: gender, race, culture



J D College of Engineering
& Management, Nagpur
An Autonomous Institute

Sources of bias: gender, race, culture



J D College of Engineering
& Management, Nagpur
An Autonomous Institute

Sources of bias: gender, race, culture



J D College of Engineering
& Management, Nagpur
An Autonomous Institute